Multi-Currency Travel Budget Analysis

Objective: To develop a Python program that converts travel expenses from different currencies into Indian Rupees (INR), calculates conversion fees, compares total expenses with the budget, and identifies the largest expense category.

1. Computational Thinking

- Decomposition: Divide the problem into currency conversion, fee calculation, budget comparison, and expense analysis.
- Pattern Recognition: Identify repeated expense calculations for different categories.
- Abstraction: Focus on exchange rates, expenses, fees, and budget.
- Algorithm Design: Follow step-by-step calculations to generate the final travel report.

2. Algorithm

1. Start.
2. Enter the total travel budget in INR.
3. Enter the number of expense categories.
4. For each category, enter:
    - Category name
    - Expense amount
    - Currency code
    - Exchange rate
5. Convert each expense into INR.
6. Calculate the conversion fee.
7. Add the converted amount and fee.
8. Store all expenses in a list.
9. Calculate total travel expenses.
10. Compare total expenses with the budget.
11. Identify the largest expense category.
12. Display the final expense report.
13. Stop.

3. Python Program

budget = float(input("Enter travel budget in INR: "))n = int(input("Enter number of expense categories: "))expenses = []total = 0for i in range(n): print("\nExpense", i + 1) category = input("Enter category: ") amount = float(input("Enter expense amount: ")) rate = float(input("Enter exchange rate: ")) fee_percent = float(input("Enter conversion fee (%): ")) converted = amount * rate fee = converted * fee_percent / 100 final_amount = converted + fee expenses.append((category, final_amount)) total += final_amountprint("\n----- TRAVEL EXPENSE REPORT -----")for category, amount in expenses: print(category, ":", round(amount, 2), "INR")largest = max(expenses, key=lambda x: x[1])print("\nTotal Expenses:", round(total, 2), "INR")print("Travel Budget:", budget, "INR")if total > budget: print("Budget Exceeded:", round(total - budget, 2))else: print("Remaining Budget:", round(budget - total, 2))print("Largest Expense Category:", largest[0])print("Largest Expense Amount:", round(largest[1], 2), "INR")

4. Sample Input

Category	Amount	Rate (INR)	Fee
Hotel	200 USD	85	2%
Food	100 EUR	90	1%
Shopping	5000 JPY	0.60	2%
Transport	50 USD	85	1%

Budget = ₹30,000

5. Sample Output

----- TRAVEL EXPENSE REPORT -----

Hotel : 17340 INR

Food : 9090 INR

Shopping : 3060 INR

Transport : 4292.5 INR

Total Expenses: 33782.5 INR

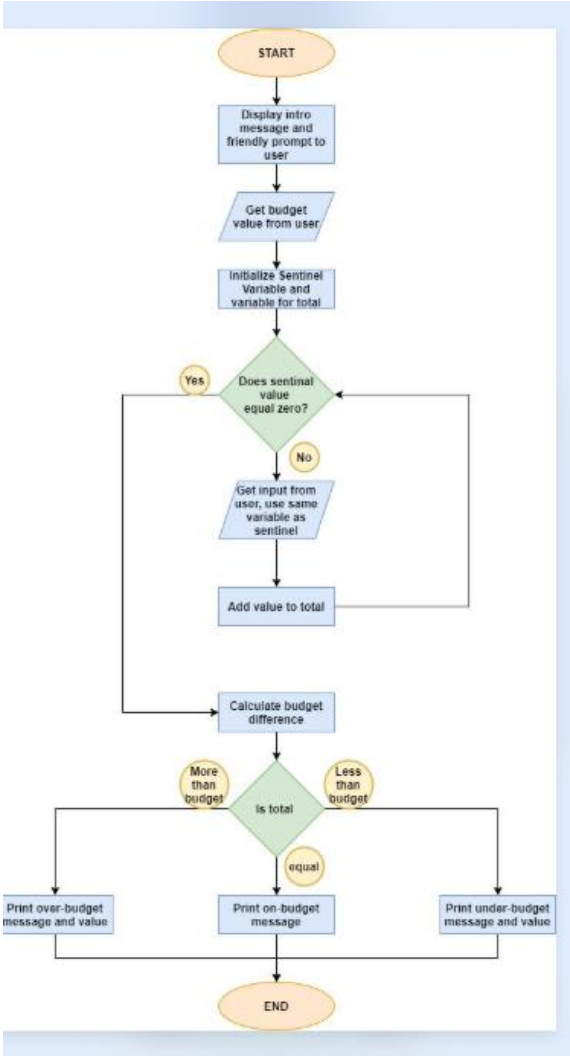
Travel Budget: 30000 INR

Budget Exceeded: 3782.5

Largest Expense Category: Hotel

Largest Expense Amount: 17340 INR

6. Flowchart



7. Concepts Used

Python Concept	**Application**
Variables	Budget, amount, rate
Data types	Integer, float, string
Operators	+, *, /, -
Conditional statements	Budget comparison
Loops	Process multiple expenses

Strings	Category and currency names
Pattern generation	Repeated expense report formatting
Lists	Store expense records
Tuples	Store category and amount
Packing	Store category and amount together
Pseudocode	Step-by-step program logic

8. Pseudocode

```
START
INPUT budget
INPUT n
total = 0

FOR each expense
   INPUT category, amount, rate, fee
   converted = amount * rate
   conversion_fee = converted * fee / 100
   final_amount = converted + conversion_fee
   STORE (category, final_amount)
   total = total + final_amount
END FOR

DISPLAY all expenses

IF total > budget
   DISPLAY budget exceeded
ELSE
   DISPLAY remaining budget
```

END IF

FIND largest expense category

DISPLAY largest category

STOP

9. Conclusion

This program helps travelers manage expenses in different currencies, calculate conversion fees, compare spending with the available budget, and identify the highest expense category using basic Python concepts.